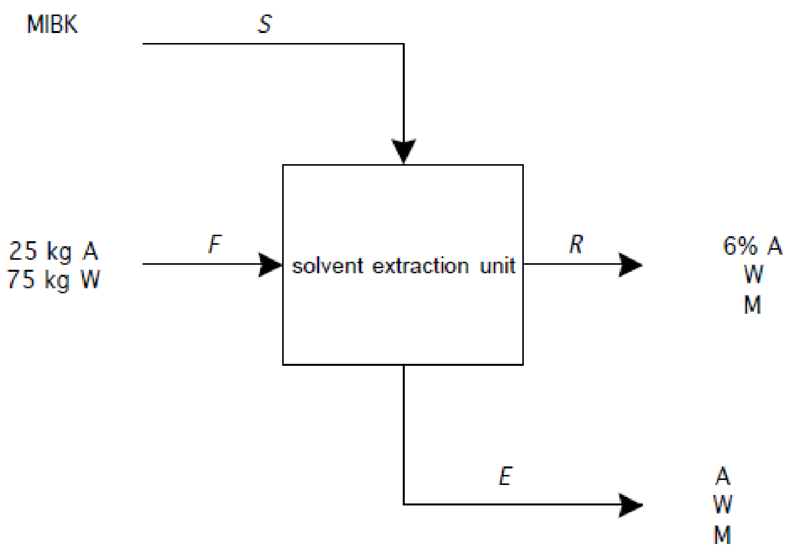
P5.40

The flow diagram is shown, using as a basis 100 kg/h feed. Compositions are given as wt%.



The raffinate *R* and extract *E* stream compositions are related by the phase equilibrium information in Figure 5.15 (and Table B.13).

Reading from the phase envelope in Figure 5.15, we see that a saturated water-rich solution containing 6 wt% acetic acid would also have ~2% MIBK and 92 wt% water. Following along an imaginary tie-line that falls in-between the two dashed tie lines in Figure 5.15, we estimate that this solution would be in equilibrium with an organic phase of 5 wt% acetic acid, 92 wt% MIBK and 3 wt% water. Thus, we find

Raffinate: 6 wt% acetic acid, 2 wt% MIBK, 92 wt% water

Extract: 5 wt% acetic acid, 92 wt% MIBK, 3 wt% water

We write material balances on all 3 components:

$$25 = 0.06\dot{m}_R + 0.05\dot{m}_E$$

$$75 = 0.92\dot{m}_R + 0.03\dot{m}_E$$

$$\dot{m}_S = 0.02\dot{m}_R + 0.92\dot{m}_E$$

We solve the first two equations simultaneously to find:

$$\begin{aligned}\dot{m}_E &= 387 \text{ kg/h} \\ \dot{m}_R &= 69 \text{ kg/h}\end{aligned} \quad \text{(at our chosen basis of 100 kg/h feed)}$$

Then we find the solvent flow rate from the 3rd equation:

$$\dot{m}_S = 357 \text{ kg/h}$$

Now we do a final check by using the total mass balance equation:

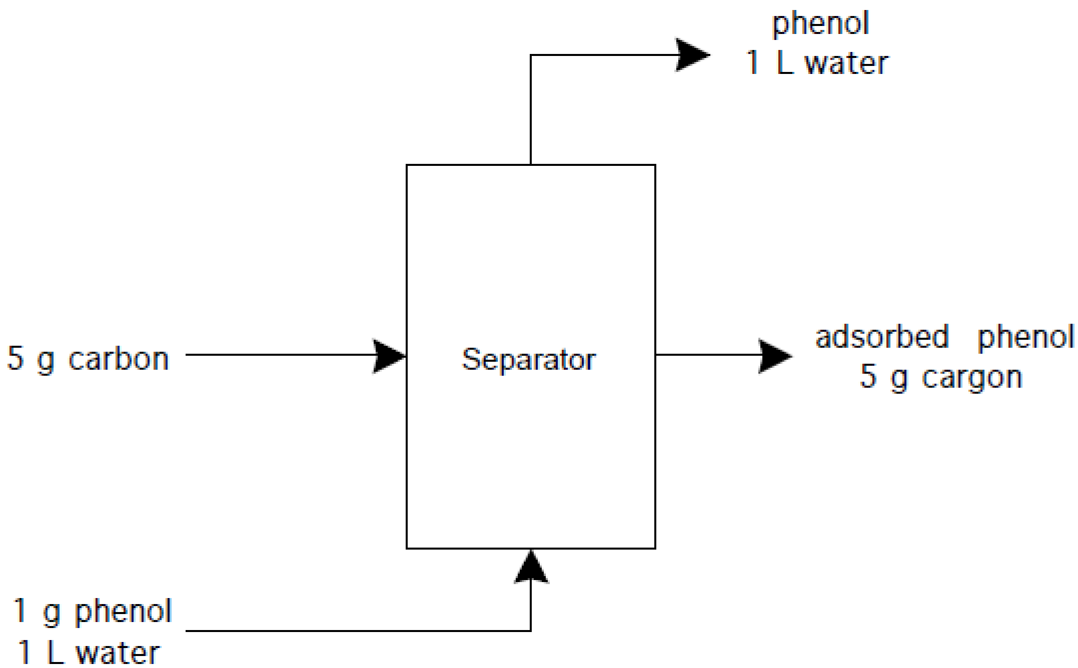
$$\dot{m}_S + \dot{m}_F = \dot{m}_E + \dot{m}_R \text{ (?)}$$

$$357 + 100 = 387 + 69 \text{ (?)}$$

$$457 = 456 \text{ (close enough)!}$$

Thus, we need 3.57 kg solvent per kg feed.

P5.41
We can draw a flow diagram as



To use the phase diagram, we need to convert to mmol: 1 g phenol (C_6H_5OH) = 10.638 mmol. Letting c_P = phenol concentration (mmol/L) in the liquid, and q_P = the phenol adsorbed (mmol/g) on the carbon, the material balance equation is simply

$$10.638 = c_P(1\text{ L}) + q_P(5\text{ g})$$

Now we need to find, by trial and error, the point (c_P, q_P) that lies on the curve and that satisfies the material balance equation. These two conditions are satisfied at $c_p \sim 0.6$ mmol/L, where $q_P = 2$ mmol/g.

$$\%\text{ removal} = \frac{2 \times 5}{10.638} \times 100\% = 95\%$$

P6.26

(a) kinetic energy change:

$$\Delta E_k = \frac{m}{2}\left(v_2{}^2 - v_1{}^2\right) = \frac{1\ \text{gmol} \times 18\ \text{g/gmol}}{2}\left((100\ \text{km/h})^2 - (0\ \text{km/h})^2\right)$$

$$= \left(90000\ \frac{\text{g km}^2}{\text{h}^2}\right) \times \left(\frac{1\ \text{kg}}{1000\ \text{g}} \times \frac{10^6\ \text{m}^2}{\text{km}^2} \times \frac{\text{h}^2}{3600^2\ \text{s}^2} \times \frac{\text{J}}{\text{kg m}^2/\text{s}^2}\right) = 6.9\ \text{J}$$

(b) potential energy change:

$$\Delta E_p = mg(h_2 - h_1) = (1\ \text{gmol} \times 18\ \text{g/gmol})\left(9.8066\ \text{m/s}^2\right)(100\ \text{m} - 0\ \text{m})$$

$$= \left(17651\ \frac{\text{g m}^2}{\text{s}^2}\right) \times \left(\frac{1\ \text{kg}}{1000\ \text{g}} \times \frac{\text{J}}{\text{kg m}^2/\text{s}^2}\right) = 17.6\ \text{J}$$

(c) internal energy change due to pressure change (from steam table):

$$\Delta U = m\Delta\hat{U} = 18\ \text{g} \times (2506.2 - 2515.5\ \text{J/g}) = -167\ \text{J}$$

(d) internal energy change due to temperature change (from liquid heat capacity):

$$\Delta U = nC_v\Delta T = 1\ \text{gmol} \times 75.4\ \text{J/gmol}\ °C \times (100 - 0°C) = 7540\ \text{J}$$

(e) internal energy change due to reaction (reverse of formation reaction)

$$\Delta\hat{U}_r \cong \Delta\hat{H}_r - RT\Delta n_r = (285.84\ \text{kJ/gmol})(1000\ \text{J/kJ}) - (8.3144\ \text{J/gmol K})(298\ K)(+0.5\ \text{gmol}) = 284{,}600\ \text{J/gm}$$

$$\Delta U_r = n\Delta\hat{U}_r = 284{,}600\ \text{J}$$

(f) energy change due to nuclear reaction

$$\Delta E = (\Delta m)c^2 = (18\ g)\left(2.99792 \times 10^8\ \text{m/s}\right)^2\left(\frac{1\ \text{kg}}{1000\ \text{g}}\right)\left(\frac{\text{J}}{\text{kg m}^2/\text{s}^2}\right) = 1.6 \times 10^{15}\ \text{J}$$